

HR Analytics Dashboard

Business Intelligence Project Report

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Project Overview

Introduction

Human Resource (HR) analytics plays a crucial role in helping organizations understand their workforce and make informed decisions that improve employee satisfaction, retention, and overall organizational performance.

This project presents an interactive HR Analytics Dashboard developed in Power BI to analyze employee attrition patterns and workforce demographics. The dashboard transforms raw HR data into meaningful visual insights, enabling stakeholders to identify trends, evaluate key performance indicators, and understand the factors influencing employee turnover.

Through interactive charts, KPI cards, and dynamic visualizations, the dashboard provides decision-makers with a comprehensive view of employee distribution, attrition rates, age groups, salary bands, education fields, and departmental performance. The project demonstrates how business intelligence tools can support strategic workforce planning and evidence-based HR decision-making.

The primary objective of this project is to convert complex HR data into clear, actionable insights that help organizations improve employee retention strategies and optimize workforce management.

Project Objectives

The HR Analytics Dashboard was developed to achieve the following objectives:

- Analyze the overall employee attrition rate within the organization.
- Examine workforce demographics, including age and gender distribution.
- Identify departments with the highest employee turnover.
- Analyze employee attrition across different age groups.
- Evaluate attrition based on education fields and qualifications.
- Examine the relationship between salary levels and employee attrition.
- Present HR metrics using interactive Power BI visualizations.
- Support HR managers and business leaders in making data-driven workforce decisions.
- Demonstrate practical skills in data visualization, dashboard design, and business intelligence using Power BI.

Dataset Description

Dataset Overview

The dataset used for this project contains employee records representing various workforce characteristics and employment information. It includes demographic details, job-related attributes, compensation data, education background, and employee attrition status.

The data was cleaned and prepared in Power BI to ensure accuracy, consistency, and effective visualization.

Key Data Fields

The dataset includes the following attributes:

- Employee ID
- Age
- Gender
- Department
- Job Role
- Education Field
- Education Level
- Monthly Income
- Attrition Status
- Years at Company
- Job Satisfaction
- Marital Status
- Business Travel
- Overtime Status

These variables provide valuable insights into workforce composition and employee turnover patterns, allowing HR professionals to identify trends and make informed business decisions.

Tools and Technologies

The following tools and technologies were used throughout the project:

Power BI Desktop

Used to build the interactive dashboard, create visualizations, design the report layout, and develop calculated measures.

Power Query

Used for data cleaning, data transformation, and preparation before visualization.

DAX (Data Analysis Expressions)

Used to create calculated measures, KPIs, and business metrics that support dashboard insights.

Data Modeling

Relationships between tables and fields were established to enable accurate analysis and interactive filtering.

Data Visualization

Various visualization techniques were used to communicate business insights effectively, including:

- KPI Cards
- Bar Charts
- Column Charts
- Donut Charts
- Tree Maps
- Slicers
- Interactive Filters

These technologies collectively enabled the creation of a dynamic and business-focused HR analytics solution.

Dashboard Features

The HR Analytics Dashboard was designed to provide users with a clear and interactive view of workforce performance through the following features:

Executive KPI Cards

The dashboard presents high-level HR metrics, including total employees, employee attrition, attrition rate, average employee age, and average monthly income, allowing decision-makers to quickly assess workforce performance.

Attrition Analysis

Interactive visuals highlight employee attrition across departments, job roles, age groups, education fields, and salary ranges, helping identify areas with higher employee turnover.

Workforce Demographics

Demographic visualizations provide insights into employee distribution by age, gender, education level, and department.

Salary Analysis

Salary-based visualizations help evaluate whether compensation levels influence employee retention and turnover.

Interactive Filtering

Users can filter dashboard results dynamically using slicers, enabling deeper exploration of workforce trends based on selected categories.

Business-Friendly Design

The dashboard follows a clean and professional layout with consistent formatting, intuitive navigation, and visually engaging charts, making it suitable for executive reporting and strategic decision-making.

Dashboard Screenshots

This section presents selected views from the HR Analytics Dashboard. These visualizations summarize key workforce metrics and provide business stakeholders with an interactive overview of employee attrition and workforce demographics.

Figure 1: Overall HR Analytics Dashboard

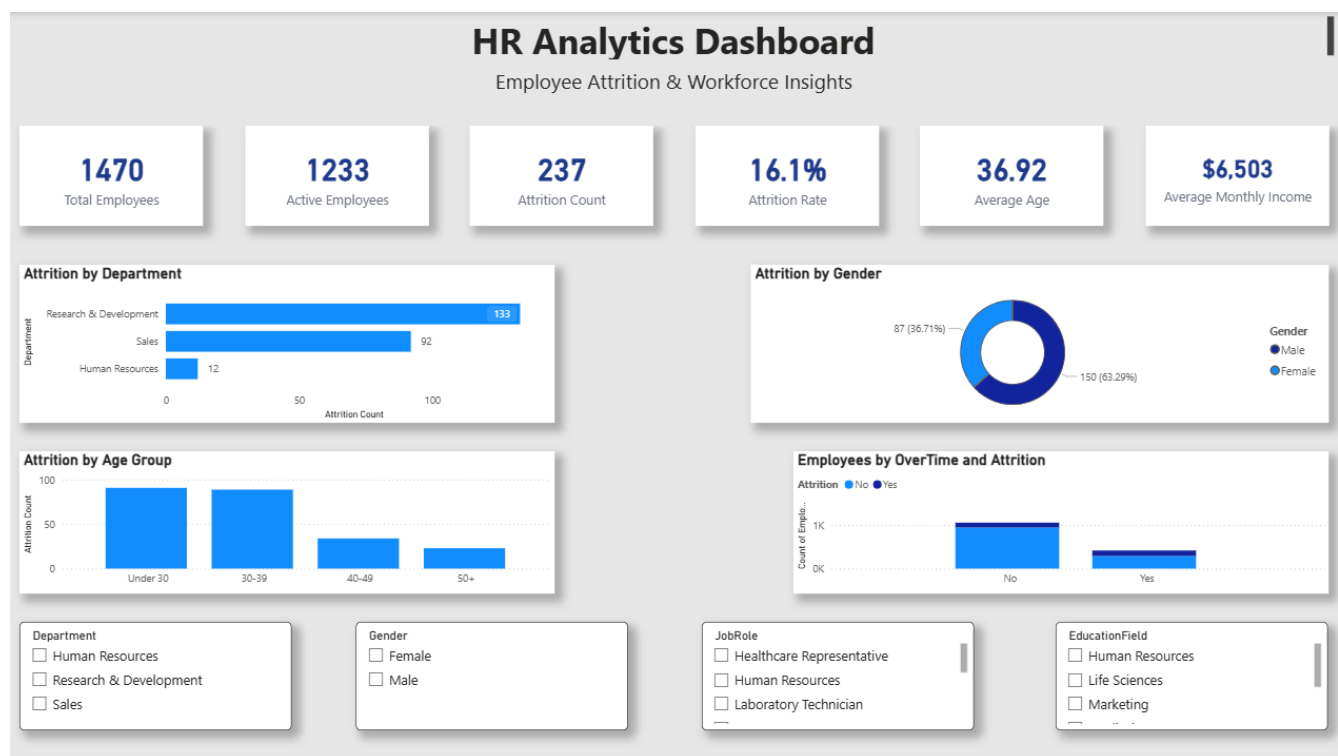


Figure 2: Employee Attrition Analysis

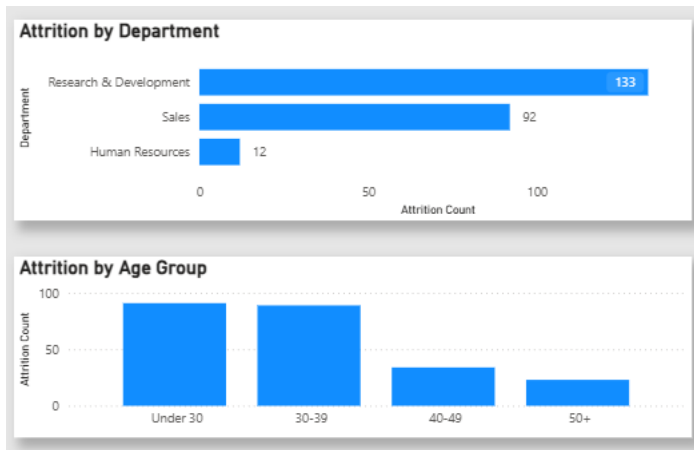


Figure 3: Workforce Demographics

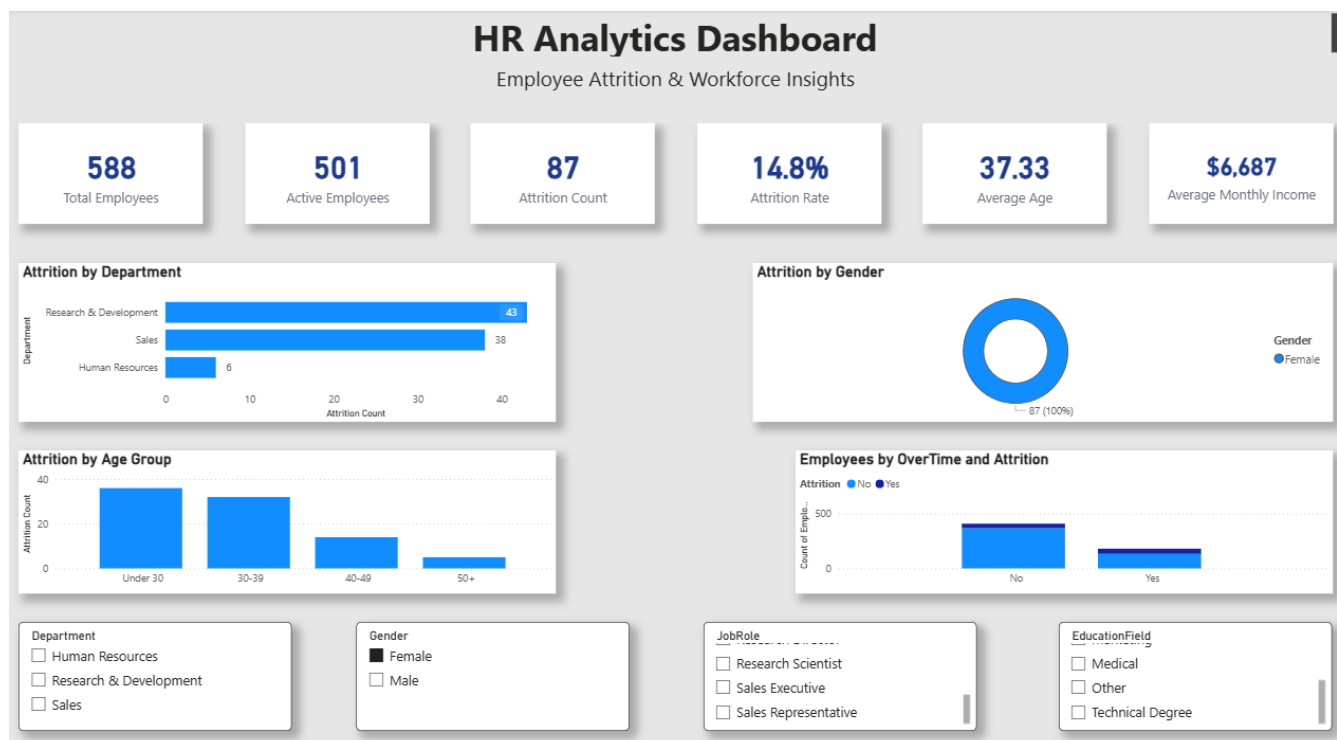
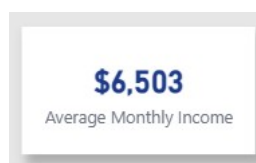


Figure 4: Salary and Income Analysis



Each visualization was designed to support quick interpretation of HR trends and enable data-driven decision-making through interactive filtering and business-focused reporting.

Key Business Insights

The dashboard reveals several important workforce trends that can support strategic HR decision-making.

Employee Attrition

The dashboard identifies the overall employee attrition level, allowing management to assess workforce stability and monitor employee retention over time.

Department Performance

Analysis of departmental attrition highlights areas experiencing higher employee turnover, enabling HR teams to prioritize retention initiatives where they are needed most.

Age Distribution

The dashboard illustrates employee attrition across different age groups, helping identify whether early-career, mid-career, or experienced employees are more likely to leave the organization.

Education Analysis

Employee attrition is analyzed across various education fields, providing insight into whether certain academic backgrounds are associated with higher turnover.

Salary Analysis

Salary-based visualizations indicate how employee turnover varies across income levels, supporting compensation reviews and workforce planning.

Workforce Demographics

Interactive demographic analysis provides a comprehensive understanding of employee distribution by gender, department, education level, and other workforce characteristics, enabling more informed HR planning.

Business Recommendations

Based on the dashboard analysis, the following recommendations can support improved workforce management and employee retention:

- Prioritize retention strategies within departments experiencing the highest employee attrition.
- Review compensation structures for employee groups with higher turnover, particularly where lower salary levels are associated with increased attrition.
- Strengthen employee engagement initiatives, including career development opportunities, mentorship programs, and recognition systems.
- Monitor workforce demographics regularly to identify emerging trends before they impact organizational performance.

- Utilize HR analytics dashboards as part of routine management reporting to support proactive, data-driven decision-making.
- Continue collecting and analyzing workforce data to measure the effectiveness of retention strategies over time.

Implementing these recommendations can contribute to improved employee satisfaction, reduced turnover, and more effective workforce planning.

Conclusion

The HR Analytics Dashboard demonstrates how business intelligence can transform raw workforce data into meaningful insights that support strategic decision-making. By integrating data preparation, analytical modeling, and interactive visualization within Power BI, the project provides a comprehensive view of employee attrition, workforce demographics, compensation trends, and departmental performance.

Beyond presenting descriptive metrics, the dashboard enables HR professionals and organizational leaders to identify workforce patterns, monitor key performance indicators, and make evidence-based decisions that improve employee retention and organizational effectiveness.

From a technical perspective, this project highlights practical proficiency in Power BI, Power Query, DAX, data modeling, and data visualization. It also demonstrates the ability to communicate analytical findings in a clear and business-focused manner, bridging the gap between data analysis and executive decision-making.

Overall, the project showcases how data analytics can deliver measurable business value by converting complex HR data into actionable insights that support long-term workforce planning and continuous organizational improvement.

About the Author

Damilola Aderemilekun Adegboye

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Damilola is passionate about transforming raw data into actionable business insights through analytics and visualization. His work focuses on designing interactive dashboards, uncovering meaningful trends, and supporting data-driven decision-making across business functions.

This project reflects his practical experience in Power BI, Python, SQL, data visualization, business intelligence, and analytical storytelling. He is committed to continuous learning and enjoys building portfolio projects that solve real-world business problems while demonstrating technical and analytical capabilities.

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